REMARKS ON THE COMPLEX STRUCTURES ON $\mathbb{P}^3$ AND $S^2 \times S^4$

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Abstract. Assuming the validity of the recently proposed *"A compact complex threefold fibred by tori over the projective line, and the six-sphere"*, we construct an exotic complex structure on $\mathbb{P}^3$, distinct from the point-blowup structures of Huckleberry, Kebekus, and Peternell. Then we perform an Atiyah flop to produce a complex structure on the standard smooth manifold $S^2 \times S^4$.

Contents

1. Introduction

After the case of S^4 had been treated by Ehresmann and Hopf [3, 4], Borel and Serre, in 1953, proved that a positive-dimensional sphere S^n admits an almost complex structure if and only if $n = 2$ or $n = 6$ [2, §15, Proposition 15.1]. Subsequently, in 1965, Sutherland [8, Theorem 3.1] showed that a product of even-dimensional spheres is almost complex exactly when it is a product of copies of

$$S^2, \qquad S^6, \qquad S^2 \times S^4.$$

Recently, a compact complex threefold X_{sph} was constructed in [7, Construction 6.1 and Theorem 6.2] and it is proved there that

$$X_{\mathrm{sph}} \cong_{\mathrm{diff}} S^6$$

as an oriented smooth manifold [7, Theorem 8.1]. The aim of this paper is to prove that $S^2 \times S^4$ has a complex structure, see theorem 1.2. Consequently, subject to the result of [7], every product of even-dimensional spheres which admits an almost complex structure also admits a complex structure.

Now we state our construction. [7] also proves that $\mathrm{Aut}^0(X_{\mathrm{sph}}) \cong \mathbb{C}^*$ [7, Proposition 9.23], and the fixed locus of $\mathrm{Aut}^0(X_{\mathrm{sph}})$ is a rational curve

$$C \cong \mathbb{P}^1,$$

where the two complex normal weights are $+1$ and -1 [7, Proposition 9.24]; every nontrivial finite subgroup has the same fixed locus [7, Remark 9.25].

It is useful to recall heuristically how this fixed curve sits in the singular fibre W. The normalization of W is the degree-six del Pezzo surface dP_6, whose toric boundary is a hexagon of (-1)-curves. W is obtained by identifying each side with its opposite side, which produces three rational double curves

$$C, \qquad D_-, \qquad D'_-.$$

These three curves pass through two triple points Q and Q' [7, Theorem 4.5 and Proposition 4.6]. The $\mathrm{Aut}^0(X_{\mathrm{sph}})$-action restricted on W is the one-parameter subgroup associated with one edge direction. The double curve in that direction is fixed pointwise; the other two double curves are preserved and the $\mathbb{C}^*$-action on each curve is the rotation fixing P and Q. On either of them the complement of P and Q is a single free $\mathbb{C}^*$-orbit [7, Proposition 9.24 and Remark 9.25]. Thus the local picture at either triple point may be visualized as

$$W = \{xyz = 0\}, \qquad \lambda \cdot (x, y, z) = (\lambda x, \lambda^{-1} y, z).$$

The z-axis is the fixed double curve, while the other two coordinate axes are invariant curves along which the action moves points toward one triple point or the other.

As proved in Proposition 5.2,

$$\widetilde{X} := \mathrm{Bl}_C(X_{\mathrm{sph}})$$

is a complex manifold diffeomorphic to $Q_3 := \{[z_0 : \cdots : z_4] \in \mathbb{P}^4 \mid z_0^2 + \cdots + z_4^2 = 0\}$. We will show in Proposition 5.2 that $\widetilde{X}$ is not biholomorphic to Q_3.

Let μ_2 be the (unique) subgroup of $\mathrm{Aut}^0(X_{\mathrm{sph}}) \cong \mathbb{C}^*$ of order 2, and $\iota \in \mu_2$ be the nontrivial element. The action of ι on both normal directions to C is multiplication by -1. Blowing up the fixed curve and taking the quotient gives a smooth complex manifold

$$X_- := \mathrm{Bl}_C(X_{\mathrm{sph}})/\mu_2.$$

We shall prove that X_- is diffeomorphic to $\mathbb{P}^3$. On the other hand, the construction of Huckleberry, Kebekus, and Peternell [5], when applied to X_{sph}, produces an exotic complex structure on $\mathbb{P}^3$ by blowing up a point of X_{sph}. The following theorem distinguishes our complex structure from theirs.

Theorem 1.1. *The complex manifold X_- is orientation-preservingly diffeomorphic to $\mathbb{P}^3$. Moreover, it is neither biholomorphic nor deformation equivalent to the standard complex structure on $\mathbb{P}^3$ and the point-blowup complex structures on $\mathbb{P}^3$ obtained from X_{sph}.*

Let C_- be the image of $(\mathrm{Bl}_C)_*^{-1}(D_-)$ in X_-. Then C_- is a smooth rational curve with normal bundle $\mathcal{O}(-1) \oplus \mathcal{O}(-1)$ (lemma 3.1). Thus, we can apply an Atiyah flop to C_- to obtain a complex manifold

$$X_+ := \mathrm{Flop}_{C_-}(X_-).$$

Theorem 1.2. *The complex manifold X_+ is diffeomorphic to $S^2 \times S^4$.*

Corollary 1.3. *If $X \cong_{\mathrm{diff}} \prod_i S^{2n_i}$ is a product of even-dimensional spheres, then X admits an almost complex structure if and only if X admits a complex structure.*

Remark 1.4. *Our construction can be summarized by the following diagram:*

$$\begin{tikzcd}
 & Q_3 \cong_{\mathrm{diff}} \widetilde{X} \arrow[dl, "\mathrm{Bl}_C"'] \arrow[dr, "/\mu_2"] & & \widehat{X} \arrow[dl, "\mathrm{Bl}_{C_-}"'] \arrow[dr, "\mathrm{Bl}_{C_+}"] & \\
S^6 \cong_{\mathrm{diff}} X_{\mathrm{sph}} & & \mathbb{P}^3 \cong_{\mathrm{diff}} X_- \arrow[rr, dashed, "\text{Atiyah flop}"] & & X_+ \cong_{\mathrm{diff}} S^2 \times S^4
\end{tikzcd}$$

The strategy of proof. By Wall's classification [9], a compact orientable simply connected spin six-dimensional differentiable manifold Y with torsion-free cohomology groups can be distinguished by the cubic form $H^2(Y) \to H^6(Y) : x \mapsto x \cup x \cup x$ and the first Pontryagin functional $p_Y(x) := \langle p_1(Y) \cup x, [Y] \rangle$. Hence, we only need to compute these topological invariants of X_- and X_+ and compare them to those of $\mathbb{P}^3$ and $S^2 \times S^4$ respectively.

To show that the complex structure obtained is an exotic one, we compute the first Chern class of $\widetilde{X}$ (or X_-) and compare them to the first Chern class of the standard complex structures on Q_3 (or $\mathbb{P}^3$ and exotic $\mathbb{P}^3$ obtained by blowing up one point on X_{sph} respectively).

The paper is organized as follows.

- In Section 2, we construct X_- as the μ_2-quotient of the blow-up of X_{sph} along the fixed curve C, and determine its fundamental group and integral homology.

- In Section 3, we identify the distinguished curve $C_- \subset X_-$, define X_+ by performing the Atiyah flop along C_-, and transport the topological calculations across the common blow-up.
- In Section 4, we compute the cubic form, the second Stiefel–Whitney class, and the first Pontryagin functional for both X_- and X_+.
- In Section 5, we apply Wall's classification theorem on certain six-manifolds. We first identify the underlying oriented smooth manifold of X_- with $\mathbb{P}^3$, and show that its complex structure is biholomorphically distinct from both the standard structure and the structure arising from the point-blow-up construction. We then identify X_+ with $S^2 \times S^4$, thereby obtaining the asserted complex structure on $S^2 \times S^4$.
- In Section 6, we give some further discussions.

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Statement on the use of AI. At first, GPT 5.6 Sol used the fibration method similar to [7] to construct a complex manifold X fibreed over $\mathbb{P}^1$ and claimed that it is homeomorphic to $S^2 \times S^4$. However, the proof provided by GPT 5.6 Sol is cumbersome, incomplete, and misses an essential lemma. Then by comparing the singular fibre of X with the singular fibres of X_{sph}, we discovered that that X is likely to be obtained by the blowup-quotient-flop process introduced in this paper, which led to the main theorem 1.2 of this article. Wall's classification result is also introduced to the authors by GPT 5.6 Sol.

The proofs of lemma 2.3, lemma 3.1, proposition 4.6 are first generated by GPT 5.6 Sol, but it is checked and rewrote by the authors.

This article is also polished by GPT 5.6 Sol on grammars, language, notations, conventions, and some graphs.

Notations and conventions. Unless otherwise stated, all manifolds are smooth, connected, and oriented, and all homology and cohomology groups have integral coefficients. For a complex manifold Y, we write $c_i(Y) := c_i(TY)$, and similarly for $p_i(Y)$ and $w_i(Y)$. If $Z \subseteq Y$ is a smooth complex submanifold, then $\mathcal{N}_{Z/Y}$ denotes its holomorphic normal bundle and $\mathrm{Bl}_Z(Y)$ denotes the blow-up of Y along Z.

If D is a Cartier divisor on Y, then $[D] := c_1(\mathcal{O}_Y(D)) \in H^2(Y)$, whereas $[Y]$ denotes the fundamental class of an oriented closed manifold. Numerical expressions such as D^3 mean $\langle [D]^3, [Y] \rangle$. The notation $\cong_{\mathrm{diff}}$ denotes an orientation-preserving diffeomorphism, and $\mathbb{P}^n$ denotes complex projective n-space.

The global geometric notation is summarized in the following table; symbols used only within a single calculation are defined at their first occurrence.

Notation	Meaning
X_{sph}	the compact complex threefold serving as the source of the construction;
$W,\ E_0,\ \nu$	the cusp fibre, its normalization $E_0 \simeq dP_6$, and the normalization map $\nu\colon E_0 \to W$;
$C,\ D_-,\ D'_-,\ Q,\ Q'$	the fixed double curve, the other two double curves of W, and the two triple points;
$\mu_2 = \{1, \iota\}$	the order-two subgroup of $\mathrm{Aut}^0(X_{\mathrm{sph}}) \cong \mathbb{C}^*$;
$\widetilde{X},\ E$	$\widetilde{X} := \mathrm{Bl}_C(X_{\mathrm{sph}})$ and its exceptional divisor;
$q,\ X_-,\ F$	the quotient map $q\colon \widetilde{X} \to X_-$ and its branch divisor $F = q(E)$;
$\widetilde{D}_-,\ C_-$	the strict transform of D_- in $\widetilde{X}$ and its image $C_- = q(\widetilde{D}_-)$;
$\widehat{X},\ R$	$\widehat{X} := \mathrm{Bl}_{C_-}(X_-)$ and its exceptional divisor;
$X_+,\ C_+$	the Atiyah flop of X_- and the image of R in X_+;
$\mathcal{L},\ \ell$	the branch half-line bundle on X_- and its class $\ell = c_1(\mathcal{L})$;
$\mathcal{L}_+,\ \ell_+$	the corresponding line bundle on X_+ and its class $\ell_+ = c_1(\mathcal{L}_+)$;
Q_3	the smooth quadric threefold in $\mathbb{P}^4$.

2. The topology of X_-

2.1. **The construction of X_-.** Near a point of C there are holomorphic coordinates (z, x, y), with z tangent to C, in which $\lambda \in \mathrm{Aut}^0(X_{\mathrm{sph}}) \cong \mathbb{C}^*$ acts as

$$\lambda \cdot (z, x, y) = (z, \lambda x, \lambda^{-1} y).$$

Hence, the element $\iota = -1 \in \mathbb{C}^*$ acts as $\iota \cdot (z, x, y) = (z, -x, -y)$. Let

$$\mathrm{Bl}_C \colon \widetilde{X} = \mathrm{Bl}_C(X_{\mathrm{sph}}) \longrightarrow X_{\mathrm{sph}}$$

be the blow-up, with exceptional divisor E. In the blow-up chart $y = xt$, the lifted involution $\widetilde{\iota}$ is

$$\widetilde{\iota} \cdot (z, x, t) = (z, -x, t).$$

Its quotient has smooth coordinates $(z, s = x^2, t)$. Hence, the action of $\widetilde{\iota}$ fixes E pointwise and the quotient map

$$q \colon \widetilde{X} \longrightarrow X_-$$

is a holomorphic double cover branched along the smooth divisor

$$F := q(E).$$

2.2. **The homology group of X_-.** We now compute the homology of X_-. Let

$$M := X_{\mathrm{sph}} \setminus C, \qquad A := M/\mu_2 \cong X_- \setminus F.$$

The homology groups of M can be obtained by Alexander duality. Since μ_2 acts freely on M, we can compute the homology groups of A by the Cartan-Leray spectral sequence. Let B be an open tubular neighborhood of the branch divisor $F \cong \mathbb{P}^1 \times \mathbb{P}^1$ in X_-. The homology of B is clear, as B is homotopy equivalent to $S^2 \times S^2$. Then we can show $A \cap B \simeq S^2 \times \mathbb{R}P^3$, $A \cup B = X_-$ and use the MV sequence to compute the homology of X_-.

Lemma 2.1. *The manifold $M = X_{\mathrm{sph}} \setminus C$ is simply connected and*

$$H_k(M) = \begin{cases} \mathbb{Z}, & k = 0, 3, \\ 0, & \textit{otherwise}. \end{cases}$$

Moreover, ι acts trivially on $H_3(M)$.

Proof. Alexander duality gives

$$\widetilde{H}_k(M) \cong \widetilde{H}^{5-k}(S^2),$$

which gives the desired homology. A loop in M bounds a disc in S^6. After a relative general-position perturbation, the disc is disjoint from C, since $\dim D^2 + \dim C = 4 < 6$. Thus $\pi_1(M) = 0$.

The generator of $H_3(M)$ is represented by a meridian S^3 around a point in C. The involution ι restricts to the antipodal map on this S^3, whose degree is $(-1)^{3+1} = +1$. Hence, the action of ι on $H_3(M)$ is trivial. □

Proposition 2.2. *The cohomology groups of A are*

$$H^k(A) = \begin{cases} \mathbb{Z}, & k = 0, 3, \\ \mathbb{Z}/2\mathbb{Z}, & k = 2, \\ 0, & \textit{otherwise}, \end{cases}$$

and the homology groups of A are

$$H_k(A) = \begin{cases} \mathbb{Z}, & k = 0, 3, \\ \mathbb{Z}/2\mathbb{Z}, & k = 1, \\ 0, & \textit{otherwise}. \end{cases}$$

Proof. For a free cellular action of a discrete group G on a CW complex X, the cohomological Cartan–Leray spectral sequence is

$$E_2^{p,q} = H^p(G; H^q(X)) \Longrightarrow H^{p+q}(X/G).$$

Here $H^p(G; -)$ denotes group cohomology. Apply this with $G = \mathbb{Z}/2\mathbb{Z}$ and $X = M$. We obtain

$$E_2^{p,q} = H^p(\mathbb{Z}/2\mathbb{Z}; H^q(M)) \Longrightarrow H^{p+q}(A).$$

Only the rows $q = 0, 3$ occur, and both coefficient modules are trivial by lemma 2.1. The group cohomology $H^*(\mathbb{Z}/2\mathbb{Z}; \mathbb{Z})$ with trivial module structure can be represented as

$$H^*(\mathbb{Z}/2\mathbb{Z}; \mathbb{Z}) = \mathbb{Z}[u]/(2u), \qquad \deg u = 2,$$

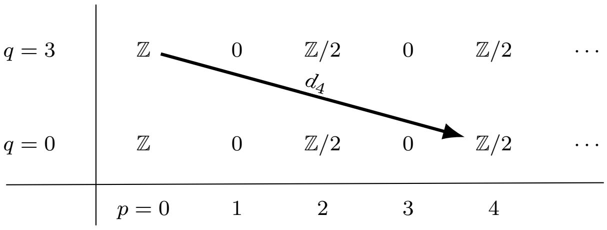
i.e.,

$$H^p(\mathbb{Z}/2\mathbb{Z}; \mathbb{Z}) = \begin{cases} \mathbb{Z}, & p = 0, \\ \mathbb{Z}/2\mathbb{Z}, & p > 0 \text{ even}, \\ 0, & p \text{ odd}. \end{cases}$$

Let x generate $H^3(M)$. Symbolically, the two nonzero rows are

$$E_2^{*,*} = \mathbb{Z}[u]/(2u) \oplus x\,\mathbb{Z}[u]/(2u), \qquad \deg u = (2, 0), \qquad \deg x = (0, 3).$$

The differential operators in the spectral sequence are $d_r \colon E_r^{p,q} \longrightarrow E_r^{p+r,q-r+1}$. The only possible nonzero differential is therefore $d_4 \colon E_4^{p,3} \longrightarrow E_4^{p+4,0}$.



We claim that d_4 at $p = 0$ is a nonzero homomorphism

$$\mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} : d_4(x) = u^2.$$

If $d_4(x) = 0$, multiplicativity would give

$$d_4(u^k x) = u^k d_4(x) = 0$$

for every k. Hence, all the classes u^k would survive in arbitrarily large total degrees. This is impossible because $H^k(A)$ must be 0 when $k > 6$. Thus $d_4(x) = u^2$. Multiplicativity gives

$$d_4(u^p x) = u^{p+2}.$$

For every $p \geq 1$, this is an isomorphism $E_4^{p,3} \to E_4^{p+4,0} : \mathbb{Z}/2 \to \mathbb{Z}/2$. It follows that u survives in degree 2, every u^p with $p \geq 2$ is killed, and every positive-column class in the upper row is killed. The surviving subgroup at $(p, q) = (0, 3)$ is

$$\ker(\mathbb{Z} \xrightarrow{\text{mod } 2} \mathbb{Z}/2) = 2\mathbb{Z} \cong \mathbb{Z}.$$

Thus the only nonzero E_∞-terms are

$$E_\infty^{0,0} = \mathbb{Z}, \qquad E_\infty^{2,0} = \mathbb{Z}/2\mathbb{Z}, \qquad E_\infty^{0,3} = 2\mathbb{Z} \cong \mathbb{Z}.$$

This proves the cohomology statement. Notice that the surviving class in degree three restricts to twice a generator of $H^3(M)$.

We finally recover homology by universal coefficient theorem. Since M is a double cover of A, we have $\pi_1(A) \cong \mathbb{Z}/2\mathbb{Z}$, and so

$$H_1(A) = \mathbb{Z}/2\mathbb{Z}.$$

The universal coefficient sequence in degree two is

$$0 \longrightarrow \mathrm{Ext}(H_1(A), \mathbb{Z}) \longrightarrow H^2(A) \longrightarrow \mathrm{Hom}(H_2(A), \mathbb{Z}) \longrightarrow 0.$$

The first two groups are both $\mathbb{Z}/2\mathbb{Z}$, so $\mathrm{Hom}(H_2(A),\mathbb{Z})=0$. The degree-three and degree-four universal coefficient sequences show respectively that $\mathrm{Ext}(H_2(A),\mathbb{Z})=0$ and that $H_3(A)$ has no torsion. As all homology groups are finitely generated, this forces

$$H_2(A)=0, \qquad H_3(A)=\mathbb{Z}.$$

Continuing in degrees four, five, and six gives

$$H_4(A)=H_5(A)=0$$

and shows that $H_6(A)$ is torsion. Since A is homotopy equivalent to a compact, connected, oriented six-manifold with nonempty boundary, $H_6(A)=0$. This proves the homology table. □

Next, we compute the homology groups of B and the intersection $A\cap B$.

By [7, Proposition 9.24 and Remark 9.25], the fixed locus of ι is a smooth curve $C=X_{\mathrm{sph}}^{\iota}\cong\mathbb{P}^1$, and ι acts by $-I$ on the normal bundle $\mathcal{N}_{C/X_{\mathrm{sph}}}$.

Lemma 2.3. *We have $\mathcal{N}_{C/X_{\mathrm{sph}}}\cong\mathcal{O}(-1)\oplus\mathcal{O}(-1)$. In particular, as an oriented real rank-four bundle, the normal bundle $\mathcal{N}_{C/X_{\mathrm{sph}}}\to C\cong_{\mathrm{diff}}S^2$ is trivial.*

Proof. Following the notation in [7, Proposition 9.24], let

$$\nu\colon E_0\simeq dP_6\longrightarrow W$$

be the normalization of the cusp fibre. By [7, Lemma 4.2(iv) and Proposition 4.6(ii)–(iii)], the two branches over C are opposite boundary curves $C_v, C_{-v}\subseteq E_0$; each map isomorphically to C and is a (-1)-curve. Hence

$$\mathcal{N}_{C_v/E_0}\cong\mathcal{N}_{C_{-v}/E_0}\cong\mathcal{O}_{\mathbb{P}^1}(-1).$$

The normal-crossings equations $xy=0$ along the smooth part of C and $xyz=0$ at its two triple points [7, Theorem 4.5(d)] show that these two branch-normal directions extend across the triple points and give an exact sequence

$$0\longrightarrow\mathcal{O}(-1)\longrightarrow\mathcal{N}_{C/X_{\mathrm{sph}}}\longrightarrow\mathcal{O}(-1)\longrightarrow 0.$$

This sequence splits because $\mathrm{Ext}^1(\mathcal{O}(-1),\mathcal{O}(-1))=H^1(\mathbb{P}^1,\mathcal{O})=0$, proving the holomorphic assertion. Finally, the underlying real bundle has $w_2=c_1(\mathcal{N}_{C/X_{\mathrm{sph}}})\bmod 2=(-2)\bmod 2=0$, and an oriented rank-four bundle over S^2 with vanishing w_2 is trivial. □

Hence, $F\cong E\cong\mathbb{P}^1\times\mathbb{P}^1$. So $\pi_1(B)=0$ and

$$H_k(B)=\begin{cases}\mathbb{Z}, & k=0,4,\\ \mathbb{Z}^2, & k=2,\\ 0, & \text{otherwise.}\end{cases}$$

Choose an open neighborhood $N(C)$ of C in X_{sph} such that

$$N(C)\cong_{\mathrm{diff}}S^2\times D^4, \qquad \iota(c,v)=(c,-v).$$

Then $N(C)\setminus C$ can be identified with a neighborhood of E in $\widetilde{X}$ minus E. Since $\partial N(C)\cong_{\mathrm{diff}}S^2\times S^3$ and ι acts on the S^3 factor by the antipodal map, we have

$$P:=\partial\mathcal{N}_{F/X_-}\cong_{\mathrm{diff}}\partial\mathcal{N}_{C/X_{\mathrm{sph}}}/\mu_2\cong_{\mathrm{diff}}S^2\times\mathbb{R}P^3.$$

Note that $A\cap B\simeq P$, the Künneth theorem gives

$$H_k(A\cap B)\cong H_k(P)=\begin{cases}\mathbb{Z}, & k=0,2,5,\\ \mathbb{Z}/2\mathbb{Z}, & k=1,\\ \mathbb{Z}\oplus\mathbb{Z}/2\mathbb{Z}, & k=3,\\ 0, & k=4.\end{cases}$$

Lemma 2.4. *The inclusion $P\hookrightarrow A$ induces:*

(1) an isomorphism on fundamental groups

$$\pi_1(P) \xrightarrow{\sim} \pi_1(A),$$

and hence an isomorphism

$$H_1(P) \xrightarrow{\sim} H_1(A);$$

(2) an isomorphism from the free summand of $H_3(P)$ onto $H_3(A)$.

Proof. A generator of $\pi_1(\mathbb{R}P^3)$ lifts to a path in S^3 from v to $-v$, and therefore represents the nontrivial deck transformation of $M \to A$. This proves the first assertion.

For the second, let

$$\alpha = [\{c\} \times \mathbb{R}P^3]$$

generate the free summand of $H_3(P)$, let a generate $H_3(A)$, and let g be the meridian generator of $H_3(M)$. Write

$$i_*\alpha = ka, \qquad \mathrm{tr}(a) = mg.$$

Transfer is natural for the square of double covers

$$\begin{array}{ccc} S^2 \times S^3 & \longrightarrow & M \\ \downarrow & & \downarrow \\ S^2 \times \mathbb{R}P^3 & \longrightarrow & A. \end{array}$$

The transfer of α is the S^3-fibre, which maps to $\pm g$ by the meridian isomorphism. Hence

$$km = \pm 1.$$

Therefore $k = \pm 1$, proving the assertion. $\square$

Lemma 2.5. *The map*

$$H_2(P) \longrightarrow H_2(B)$$

is injective with torsion-free rank-one cokernel.

Proof. The neighborhood B is the real rank-two disc bundle of the normal complex line bundle of F, with sphere boundary P. By the Thom isomorphism,

$$H_3(B, P) \cong H_1(F) = 0, \qquad H_2(B, P) \cong H_0(F) = \mathbb{Z}.$$

The long exact sequence of the pair contains

$$0 \longrightarrow H_2(P) \longrightarrow H_2(B) \longrightarrow \mathbb{Z} \longrightarrow H_1(P) \cong \mathbb{Z}/2\mathbb{Z} \longrightarrow 0.$$

The map $\mathbb{Z} \to H_1(P) \cong \mathbb{Z}/2\mathbb{Z}$ is surjective, so its kernel is $2\mathbb{Z}$. It follows that the first map is injective and its cokernel is isomorphic to $2\mathbb{Z} \cong \mathbb{Z}$, hence is torsion-free. $\square$

Now we apply the MV-sequence to $X_- = A \cup B$.

Proposition 2.6. *The manifold $X_- = A \cup B$ is simply connected and*

$$H_k(X_-) = \begin{cases} \mathbb{Z}, & k = 0, 2, 4, 6, \\ 0, & k = 1, 3, 5. \end{cases}$$

Proof. The van Kampen theorem gives

$$\pi_1(X_-) \cong \pi_1(A) *_{\pi_1(P)} \pi_1(B) = 1,$$

because $\pi_1(P) \to \pi_1(A)$ is an isomorphism and $\pi_1(B) = 1$.

Consider the MV-sequence

$$\begin{tikzcd} \cdots \arrow[r] & H_3(A \cap B) \arrow[r] & H_3(A) \oplus H_3(B) \arrow[r] & H_3(X_-) \arrow[dll, "\partial"] \\ & H_2(A \cap B) \arrow[r] & H_2(A) \oplus H_2(B) \arrow[r] & H_2(X_-) \arrow[dll, "\partial"] \\ & H_1(A \cap B) \arrow[r] & H_1(A) \oplus H_1(B) \arrow[r] & \cdots \end{tikzcd} \tag{1}$$

Substituting the groups already calculated, this becomes

$$\mathbb{Z} \oplus \mathbb{Z}/2 \xrightarrow{(\pm 1, 0)} \mathbb{Z} \longrightarrow H_3(X_-) \longrightarrow \mathbb{Z} \hookrightarrow \mathbb{Z}^2 \longrightarrow H_2(X_-) \longrightarrow \mathbb{Z}/2 \xrightarrow{\sim} \mathbb{Z}/2.$$

The first arrow is surjective by lemma 2.4; its torsion summand necessarily maps to zero. Hence $H_3(X_-)$ injects into $H_2(P) = \mathbb{Z}$. Its image is the kernel of the displayed injection $\mathbb{Z} \hookrightarrow \mathbb{Z}^2$, and is therefore zero. Thus $H_3(X_-) = 0$.

At the other end, the last arrow is an isomorphism, so $\mathbb{Z}^2 \to H_2(X_-)$ is surjective. Its kernel is the image of $H_2(P)$, and lemma 2.5 identifies the cokernel of that image with $\mathbb{Z}$. Therefore

$$H_3(X_-) = 0, \qquad H_2(X_-) = \mathbb{Z}.$$

Simple connectivity gives $H_1(X_-) = 0$. Since X_- is a closed connected oriented six-manifold, $H_0(X_-) = H_6(X_-) = \mathbb{Z}$. The universal coefficient theorem gives $H^2(X_-) \cong \mathrm{Hom}(H_2(X_-), \mathbb{Z}) = \mathbb{Z}$ and $H^1(X_-) = 0$. Poincaré duality then gives

$$H_4(X_-) \cong H^2(X_-) \cong \mathbb{Z}, \qquad H_5(X_-) \cong H^1(X_-) = 0,$$

which completes the table. □

3. The topology of X_+

3.1. The construction of X_+. For the same reason as lemma 2.3, the normal bundle of the curves D_- is isomorphic to $\mathcal{O}(-1) \oplus \mathcal{O}(-1)$. Let $\widetilde{D}_- \subseteq \widetilde{X}$ be its strict transform by the blow up Bl_C. We claim that

Lemma 3.1. *The image $C_- := q(\widetilde{D}_-) \subseteq X_-$ is a smooth rational curve and*

$$\mathcal{N}_{C_-/X_-} \cong \mathcal{O}(-1) \oplus \mathcal{O}(-1).$$

Proof. Identifying $\widetilde{D}_-$ with D_- via the blow up map $b : \widetilde{X} = \mathrm{Bl}_C(X_{\mathrm{sph}}) \to X_{\mathrm{sph}}$ and blowing up the two transverse intersection points Q', Q' gives an elementary exact sequence

$$0 \longrightarrow \mathcal{N}_{\widetilde{D}_-/\widetilde{X}} \longrightarrow \mathcal{N}_{D_-/X_{\mathrm{sph}}} \longrightarrow \mathbb{C}_Q \oplus \mathbb{C}_{Q'} \longrightarrow 0.$$

According to lemma 2.3, $\mathcal{N}_{D_-/X_{\mathrm{sph}}} \cong \mathcal{O}(-1) \oplus \mathcal{O}(-1)$ .Thus the only possibilities of $\mathcal{N}_{\widetilde{D}_-/\widetilde{X}}$ are

$$\mathcal{O}(-2) \oplus \mathcal{O}(-2) \qquad \text{or} \qquad \mathcal{O}(-1) \oplus \mathcal{O}(-3). \tag{2}$$

We now exclude the second possibility. By [7, Propositions 9.23–9.24 and Remark 9.25],

$$q|_{\widetilde{D}_-} : \widetilde{D}_- \longrightarrow C_-$$

is a double cover branched at the points over Q, Q'. The quotient is smooth along C_-, and Riemann–Hurwitz gives $C_- \cong \mathbb{P}^1$. Moreover, the normal direction coordinates near $\widetilde{D}_-$ descend unchanged, so $\mathcal{N}_{\widetilde{D}_-/\widetilde{X}} \cong \rho^* \mathcal{N}_{C_-/X_-}$. Write $\mathcal{N}_{C_-/X_-} \cong \mathcal{O}(m) \oplus \mathcal{O}(n)$. Since $\deg q|_{\widetilde{D}_-} = 2$, we have that

$$\mathcal{N}_{\widetilde{D}_-/\widetilde{X}} \cong \mathcal{O}(2m) \oplus \mathcal{O}(2n).$$

Both splitting degrees upstairs are therefore even. This excludes $\mathcal{O}(-1) \oplus \mathcal{O}(-3)$ in (2), and hence

$$\mathcal{N}_{\widetilde{D}_-/\widetilde{X}} \cong \mathcal{O}_{\mathbb{P}^1}(-2) \oplus \mathcal{O}_{\mathbb{P}^1}(-2).$$

Consequently, $m = n = -1$, and the lemma follows. □

Therefore, we can apply the standard (analytic) Atiyah flop to X_-.

Definition 3.2. *Let*

$$\widehat{X} := \mathrm{Bl}_{C_-} X_- \longrightarrow X_-$$

be the blow-up, and let $R \subseteq \widehat{X}$ be its exceptional divisor. Then

$$R \cong \mathbb{P}^1 \times \mathbb{P}^1, \qquad \mathcal{O}_{\widehat{X}}(R)|_R \cong \mathcal{O}_R(-1, -1).$$

Contracting R along another ruling gives a smooth compact complex threefold X_+ with a curve C_+ (the image of R) and a morphism

$$\mathrm{Bl}_{C_+} : \widehat{X} \longrightarrow X_+.$$

We call X_+ the Atiyah flop *of X_- along C_-.*

The contraction in the definition is the standard analytic Atiyah contraction [1]. By construction, we have $X_- \setminus C_- = X_+ \setminus C_+ = \widehat{X} \setminus R$.

(3)
$$\begin{tikzcd} & \widehat{X} \arrow[dl, "\mathrm{Bl}_{C_-}"'] \arrow[dr, "\mathrm{Bl}_{C_+}"] & \\ X_- & & X_+ \end{tikzcd}$$

Both arrows are blow-ups of a smooth rational curve with normal bundle $\mathcal{O}(-1) \oplus \mathcal{O}(-1)$.

3.2. **The homology groups of X_+.** It is standard to compute the homology groups across the Atiyah flop. For completeness of the paper, we give a sketch here.

Lemma 3.3. *Let $a \colon \widetilde{Y} = \mathrm{Bl}_C Y \to Y$ be the blow-up of a complex manifold along a smooth connected complex codimension-two submanifold. Then*

$$H^k(\widetilde{Y}) \cong H^k(Y) \oplus H^{k-2}(C).$$

Proof. Let $E = \mathbb{P}(\mathcal{N}_{C/Y})$, let $\pi \colon E \to C$ be the projection, and let $j \colon E \hookrightarrow \widetilde{Y}$ be the inclusion. Excision and the Thom isomorphism compare the long exact sequences of $(Y, Y \setminus C)$ and $(\widetilde{Y}, \widetilde{Y} \setminus E)$. Together with the integral projective-bundle decomposition

$$H^*(E) = \pi^* H^*(C) \oplus \xi\, \pi^* H^{*-2}(C), \qquad \xi = c_1(\mathcal{O}_E(1)),$$

the comparison map is the identity off the exceptional locus and sends the Thom summand of C to the ξ-summand of E. Five Lemma applied to the two long exact sequences therefore gives what we want. □

Lemma 3.4. *Both the blow-up maps in the Atiyah flop induce isomorphisms*

$$\pi_1(\widehat{X}) \cong \pi_1(X_-), \qquad \pi_1(\widehat{X}) \cong \pi_1(X_+).$$

Proof. We treat $\widehat{X} \to X_-$. Let T be a tubular neighbourhood of $C_- \cong \mathbb{P}^1$ and put $U = X_- \setminus T$. The space T retracts onto C_- and is simply connected. Its boundary is homeomorphic to $S^2 \times S^3$, hence is simply connected. The inverse image of T retracts onto $R \cong \mathbb{P}^1 \times \mathbb{P}^1$, so it is also simply connected. Its boundary is the same as ∂T. Applying the van Kampen theorem proves the lemma. □

Proposition 3.5. *X_+ is simply connected and*

$$H_k(X_+) = \begin{cases} \mathbb{Z}, & k = 0, 2, 4, 6, \\ 0, & k = 1, 3, 5. \end{cases}$$

Proof. Apply lemma 3.3 to both arrows of the flop:

$$H^k(\widehat{X}) \cong H^k(X_-) \oplus H^{k-2}(\mathbb{P}^1),$$
$$H^k(\widehat{X}) \cong H^k(X_+) \oplus H^{k-2}(\mathbb{P}^1).$$

The added group is $\mathbb{Z}$ for $k = 2, 4$, and is zero in every other degree. By proposition 2.6 and integral Poincaré duality, the first decomposition gives

$$H^k(\widehat{X}) = \begin{cases} \mathbb{Z}, & k = 0, 6, \\ \mathbb{Z}^2, & k = 2, 4, \\ 0, & k = 1, 3, 5. \end{cases}$$

Comparing with the second decomposition degree by degree gives

$$H^k(X_+) = \begin{cases} \mathbb{Z}, & k = 0, 2, 4, 6, \\ 0, & k = 1, 3, 5. \end{cases}$$

For $k = 2, 4$, this cancels one free $\mathbb{Z}$-summand. The classification of finitely generated abelian groups also shows that no torsion can occur.

By lemma 3.4 and proposition 2.6,

$$\pi_1(X_+) \cong \pi_1(\widehat{X}) \cong \pi_1(X_-) = 1.$$

The universal coefficient theorem, together with integral Poincaré duality, converts the cohomology table into the stated homology table and confirms that no hidden torsion occurs. ☐

Remark 3.6. *An flop preserve the cohomology and homology groups (as additive groups), but it may change the ring structure of the cohomology ring, thus the cubic form and the first Pontryagin functional. Those invariants are computed separately below.*

4. The Wall invariants of X_- and X_+

4.1. Characteristic classes of X_-.

Let

$$\mathrm{Bl}_C \colon \widetilde{X} := \mathrm{Bl}_C(X_{\mathrm{sph}}) \longrightarrow X_{\mathrm{sph}}$$

be the blow-up of the fixed curve, and denote its exceptional divisor by E. The lifted involution $\widetilde{\iota}$ fixes E pointwise, and gives a holomorphic double covering

$$q \colon \widetilde{X} \longrightarrow X_-$$

branched along the exceptional divisor

$$F := q(E) \subseteq X_-, \qquad q^*F = 2E.$$

Note that q is finite and flat of degree two. $q_*\mathcal{O}_{\widetilde{X}}$ is locally free of rank two. Consider the trace morphism associated with the finite morphism q. We have an exact sequence

$$0 \to \ker(\mathrm{Tr}) \to q_*\mathcal{O}_{\widetilde{X}} \xrightarrow{\mathrm{Tr}} \mathcal{O}_{X_-} \to 0.$$

The trace morphism splits the inclusion of constants because $\mathrm{Tr} = \mathrm{id} + \iota^*$. Then let

$$\mathcal{L} := \ker(\mathrm{Tr})^{-1}$$

be a line bundle. The trace decomposition is

$$q_*\mathcal{O}_{\widetilde{X}} = \mathcal{O}_{X_-} \oplus \mathcal{L}^{-1}, \qquad \mathcal{L}^{-1} = \ker(\mathrm{Tr}).$$

The product of two anti-invariant sections is invariant and therefore induces a homomorphism

$$\mathcal{L}^{-1} \otimes \mathcal{L}^{-1} \longrightarrow \mathcal{O}_{X_-},$$

or equivalently a section

$$\sigma \in H^0(X_-, \mathcal{L}^{\otimes 2}).$$

On an open set $U \subset X_-$ with a frame e of $\mathcal{L}$, write $\sigma = fe^{\otimes 2}$, and let t denote the element of the trace-zero summand corresponding to e^{-1}. Then

$$q_*\mathcal{O}_{\widetilde{X}}|_U \cong \mathcal{O}_U[t]/(t^2 - f).$$

Let $F = \mathrm{div}(\sigma)$ be the branch divisor and let $E \subset \widetilde{X}$ be the ramification divisor. The section σ gives

$$\mathcal{L}^{\otimes 2} \cong \mathcal{O}_{X_-}(F).$$

Moreover, the local sections $t\, q^*e$ glue to a global section of $q^*\mathcal{L}$ whose zero divisor is E. Hence

$$q^*\mathcal{L} \cong \mathcal{O}_{\widetilde{X}}(E), \qquad q^*F = 2E. \tag{4}$$

Setting

$$\ell := c_1(\mathcal{L}) \in H^2(X_-),$$

we obtain

$$[F] := c_1(\mathcal{O}_{X_-}(F)) = 2\ell.$$

Lemma 4.1. *We have*

$$E^3 = 2, \qquad \langle \ell^3, [X_-] \rangle = 1.$$

In particular, the class ℓ is a generator of $H^2(X_-)$.

Proof. For the blow-up of a complex threefold along a smooth curve, let

$$\pi \colon E = \mathbb{P}(\mathcal{N}_{C/X_{\mathrm{sph}}}) \longrightarrow C, \qquad \eta := c_1\big(\mathcal{O}_{\mathbb{P}(\mathcal{N}_{C/X_{\mathrm{sph}}})}(-1)\big).$$

Thus, on each fibre $\pi^{-1}(c) \cong \mathbb{P}^1$, the class η has degree -1. Then $\mathcal{O}_{\widetilde{X}}(E)|_E \cong \mathcal{O}_{\mathbb{P}(\mathcal{N}_{C/X_{\mathrm{sph}}})}(-1)$ and $[E]|_E = \eta$. The push-forward formulas give

$$\pi_*(\eta) = -1, \qquad \pi_*(\eta^2) = -c_1(\mathcal{N}_{C/X_{\mathrm{sph}}}).$$

By lemma 2.3, we have $\deg c_1(\mathcal{N}_{C/X_{\mathrm{sph}}}) = -2$. Consequently,

$$E^3 = \int_E \eta^2 = -\int_C c_1(\mathcal{N}_{C/X_{\mathrm{sph}}}) = 2.$$

Note that the degree of q is two. By equation (4) we have

$$2\langle \ell^3, [X_-] \rangle = \langle q^*\ell^3, [\widetilde{X}] \rangle = E^3 = 2.$$

Hence $\ell^3 = 1$ and so ℓ is a generator. □

Now we compute the second Stiefel-Whitney class and the Pontryagin functional on X_-.

Proposition 4.2. *The first Chern class of X_- vanishes. In particular,*

$$w_2(X_-) = 0.$$

Proof. The canonical-bundle formulas for blow-up along C and for the double cover q branched in $F \in |2\mathcal{L}|$ give

$$\mathrm{Bl}_C^* K_{X_{\mathrm{sph}}} \otimes \mathcal{O}_{\widetilde{X}}(E) \cong K_{\widetilde{X}} \cong q^*(K_{X_-} \otimes \mathcal{L}).$$

Using (4), their first Chern classes give $q^*c_1(K_{X_-}) = \mathrm{Bl}_C^* c_1(K_{X_{\mathrm{sph}}}) = 0$. Therefore

$$2c_1(K_{X_-}) = q_*q^*c_1(K_{X_-}) = 0.$$

By proposition 2.6, $H^2(X_-) \cong \mathbb{Z}$ is torsion-free, so $c_1(K_{X_-}) = 0$. Hence

$$w_2(X_-) = w_2(TX_-) = c_1(-K_{X_-}) \bmod 2 = 0.$$

□

Proposition 4.3. *The first Pontryagin functional $p_{X_-}(x) = \langle p_1(X_-) \cup x, [X_-] \rangle$ of X_- satisfies $p_{X_-}(\ell) = 4$.*

Proof. Since $F \cong_{\mathrm{diff}} \mathbb{P}^1 \times \mathbb{P}^1$, the signature of F is zero. Moreover $[F] = 2\ell$ and $\ell^3 = 1$, hence

$$\langle [F]^3, [X_-] \rangle = \langle (2\ell)^3, [X_-] \rangle = 8.$$

For the oriented real two-plane bundle underlying the normal complex line bundle, $p_1(\mathcal{N}_{F/X_-}) = c_1(\mathcal{N}_{F/X_-})^2$. The self-intersection formula gives

$$c_1(\mathcal{N}_{F/X_-}) = i^*[F], \qquad i \colon F \hookrightarrow X_-,$$

and therefore

$$\Big\langle c_1(\mathcal{N}_{F/X_-})^2, [F] \Big\rangle = \langle [F]^3, [X_-] \rangle = 8.$$

The real normal sequence along F splits, so as real vector bundles $TX_-|_F \cong TF \oplus \mathcal{N}_{F/X_-}$. The Hirzebruch signature theorem on the four-manifold F therefore gives

$$\begin{aligned}
\langle p_1(X_-) \cup \ell, [X_-] \rangle &= \frac{1}{2} \langle p_1(X_-) \cup [F], [X_-] \rangle \\
&= \frac{1}{2} \langle p_1(TF), [F] \rangle + \frac{1}{2} \langle c_1(\mathcal{N}_{F/X_-})^2, [F] \rangle \\
&= \frac{3}{2}\,\mathrm{sign}(F) + \frac{1}{2} F^3 = 4.
\end{aligned}$$

□

4.2. **Characteristic classes of X_+.** Consider the flop described in equation (3). Let $R \subseteq \widehat{X}$ be the exceptional divisor. By Section 3

$$C_- \cong \mathbb{P}^1, \qquad \mathcal{N}_{C_-/X_-} \cong \mathcal{O}(-1) \oplus \mathcal{O}(-1), \qquad R \cong \mathbb{P}^1 \times \mathbb{P}^1.$$

We now compute the characteristic classes of X_+.

Since C intersects with D_- in Q, Q' transversely, the curve C_- satisfies $F \cdot C_- = 2$ and $\ell \cdot C_- = 1$. Choose the rulings of $R \cong \mathbb{P}^1 \times \mathbb{P}^1$ so that $\mathrm{Bl}_{C_-}|_R$ is the projection to the first factor and $\mathrm{Bl}_{C_+}|_R$ is the projection to the second. The line bundle $\mathcal{L}$ has degree one on C_-, and therefore

$$\mathrm{Bl}^*_{C_-}(\mathcal{L})|_R \cong \mathcal{O}_R(1,0), \qquad \mathcal{O}_{\widehat{X}}(R)|_R \cong \mathcal{O}_R(-1,-1). \tag{5}$$

For the blow up along C_+, we have

$$\mathrm{Pic}(\widehat{X}) \cong \mathrm{Bl}^*_{C_+} \mathrm{Pic}(X_+) \oplus \mathbb{Z}[\mathcal{O}_{\widehat{X}}(R)],$$

where the second coefficient is detected by degree on an exceptional fibre. Indeed, restriction defines a degree homomorphism $\mathrm{Pic}(\widehat{X}) \to \mathbb{Z}$, the bundle $\mathcal{O}_{\widehat{X}}(R)$ has degree -1, and the kernel consists precisely of pullbacks: a line bundle of degree zero restricts on $R = \mathbb{P}(\mathcal{N}_{C_+/X_+})$ as the pullback of a line bundle on C_+, and hence descends uniquely across the smooth centre. Therefore there is a unique holomorphic line bundle $\mathcal{L}_+$ on X_+ such that

$$\mathrm{Bl}^*_{C_+} \mathcal{L}_+ \cong \mathrm{Bl}^*_{C_-} \mathcal{L} \otimes \mathcal{O}_{\widehat{X}}(R). \tag{6}$$

Put

$$\ell_+ := c_1(\mathcal{L}_+).$$

Let $U = X_- \setminus C_- = X_+ \setminus C_+$ be the complement of the two flopping curves. The excision gives $H^k(X_\pm, U) \cong H^{k-4}(C_\pm)$. Hence the relative groups vanish for $k = 2, 3$, and the long exact sequences of the pairs give isomorphisms

$$H^2(X_-) \cong H^2(U) \cong H^2(X_+).$$

Restricting (6) to the common complement gives $\ell|_U = \ell_+|_U$. Thus these isomorphisms identify ℓ with ℓ_+.

Proposition 4.4. *The generator $\ell_+ \in H^2(X_+)$ satisfies*

$$\langle \ell_+^3, [X_+] \rangle = 0.$$

Proof. Choose the two rulings of $R \cong \mathbb{P}^1 \times \mathbb{P}^1$ so that $\mathrm{Bl}_{C_-}|_R$ is projection to the first factor. By equation (5), we have the intersection formulas

$$(\mathrm{Bl}^*_{C_-} \ell)^2 R = 0, \qquad \mathrm{Bl}^*_{C_-} \ell \, R^2 = -1, \qquad R^3 = 2.$$

Using equation (6) and $\ell^3 = 1$, we obtain

$$\begin{aligned}
\left\langle \ell_+^3, [X_+] \right\rangle &= \left\langle (\mathrm{Bl}^*_{C_+} \ell_+)^3, [\widehat{X}] \right\rangle \\
&= \left\langle (\mathrm{Bl}^*_{C_-} \ell + R)^3, [\widehat{X}] \right\rangle \\
&= \ell^3 + 3(\mathrm{Bl}^*_{C_-} \ell)^2 R + 3 \,\mathrm{Bl}^*_{C_-} \ell \, R^2 + R^3 \\
&= 1 + 0 - 3 + 2 = 0.
\end{aligned}$$

□

We next compute the Chern classes and Pontryagin classes after the flop.

Proposition 4.5. *The first Chern class of X_+ vanishes. In particular,*

$$w_2(X_+) = 0.$$

Proof. The canonical-bundle formulas for the two blow-ups give

$$K_{\widehat{X}} \cong \mathrm{Bl}_{C_-}^* K_{X_-} \otimes \mathcal{O}_{\widehat{X}}(R) \cong \mathrm{Bl}_{C_+}^* K_{X_+} \otimes \mathcal{O}_{\widehat{X}}(R).$$

Thus $\mathrm{Bl}_{C_-}^* c_1(K_{X_-}) = \mathrm{Bl}_{C_+}^* c_1(K_{X_+})$. The left side is zero by proposition 4.2, and pullback by a blow-up is injective in degree two by lemma 3.3. Hence $c_1(K_{X_+}) = 0$, so $c_1(TX_+) = -c_1(K_{X_+}) = 0$. Reduction modulo two gives $w_2(X_+) = 0$. □

Proposition 4.6. *The first Pontryagin functional $p_{X_+}(x) = \langle p_1(X_+) \cup x, [X_+]\rangle$ of X_+ is identically zero.*

Proof. We first compare the Euler characteristics of $\mathcal{L}$ and $\mathcal{L}_+$. Restricting equation (6) to R, we have

$$\mathrm{Bl}_{C_+}^* \mathcal{L}_+|_R \cong \mathcal{O}_R(1,0) \otimes \mathcal{O}_R(-1,-1) \cong \mathcal{O}_R(0,-1),$$

whose cohomology vanishes in every degree. The divisor exact sequence

$$0 \longrightarrow \mathrm{Bl}_{C_-}^* \mathcal{L} \longrightarrow \mathrm{Bl}_{C_-}^* \mathcal{L} \otimes \mathcal{O}_{\widehat{X}}(R) \longrightarrow \mathcal{O}_R(0,-1) \longrightarrow 0$$

therefore gives

$$\chi(\widehat{X}, \mathrm{Bl}_{C_+}^* \mathcal{L}_+) = \chi(\widehat{X}, \mathrm{Bl}_{C_-}^* \mathcal{L}).$$

For a blow-up along a smooth centre,

$$R^i \,\mathrm{Bl}_{C_+,*} \mathcal{O}_{\widehat{X}} = R^i \,\mathrm{Bl}_{C_-,*} \mathcal{O}_{\widehat{X}} = 0 \quad (i > 0),$$

and both direct images in degree zero are the structure sheaves. The projection formula consequently yields

$$\chi(X_+, \mathcal{L}_+) = \chi(X_-, \mathcal{L}).$$

For the compact complex threefold X_+, since $c_1(TX_+) = 0$, the Hirzebruch–Riemann–Roch formula gives

$$\begin{aligned}\chi(X_+, \mathcal{L}_+) &= \frac{1}{6}\langle l_+^3, [X_+]\rangle + \frac{1}{12}\langle c_2(X_+) \cup l_+, [X_+]\rangle \\ &= \frac{1}{24}\langle 4l_+^3 - p_1(X_+) \cup l_+, [X_+]\rangle,\end{aligned}$$

because $p_1 = c_1^2 - 2c_2 = -2c_2$. The same formula applies to $\chi(X_-, \mathcal{L})$, hence

$$24\chi(X_-, \mathcal{L}) = 4\left\langle \ell^3, [X_-]\right\rangle - \langle p_1(X_-) \cup \ell, [X_-]\rangle = 4 - 4 = 0$$

by lemma 4.1 and proposition 4.3. On X_+, $\ell_+^3 = 0$ by proposition 4.4. Hence

$$\langle p_1(X_+) \cup \ell_+, [X_+]\rangle = 0.$$

□

5. Proof of the main theorems

We use Wall's smooth classification theorem for spin six-manifolds [9]. Jupp established the corresponding broader classification of simply connected six-manifolds with torsion-free integral homology, including the non-spin case [6, Theorem 1 and the corollary on p. 299].

We now state precisely the form of Wall's result used in this paper.

Theorem 5.1. [9, Theorem 5] *An admissible spin Wall system in dimension six is a quadruple*

$$(r, H, \mu, p)$$

consisting of a nonnegative integer r, a finitely generated free abelian group H, a symmetric trilinear form

$$\mu\colon H \times H \times H \longrightarrow \mathbb{Z},$$

and a homomorphism $p\colon H \to \mathbb{Z}$, subject to the congruences

$$\mu(x,x,y) + \mu(x,y,y) \equiv 0 \pmod 2 \qquad \textit{for all } x, y \in H, \tag{7}$$

$$4\mu(x,x,x) - p(x) \equiv 0 \pmod{24} \qquad \textit{for all } x \in H. \tag{8}$$

For a closed, connected, oriented, simply connected spin smooth real six-manifold Y with torsion-free integral homology, set

$$H_Y := H^2(Y), \qquad r_Y := \frac{1}{2}\operatorname{rank} H^3(Y),$$

$$\mu_Y(x,y,z) := \langle x \cup y \cup z, [Y]\rangle, \qquad p_Y(x) := \langle p_1(Y) \cup x, [Y]\rangle.$$

Then r_Y is an integer and (r_Y, H_Y, μ_Y, p_Y) is admissible. The assignment

$$Y \longmapsto (r_Y, H_Y, \mu_Y, p_Y)$$

induces a bijection from orientation-preserving diffeomorphism classes of such manifolds to isomorphism classes of admissible systems. Explicitly, two such manifolds Y and Y' are orientation-preservingly diffeomorphic if and only if $r_Y = r_{Y'}$ and there is an isomorphism $\varphi\colon H_Y \to H_{Y'}$ satisfying

$$\mu_{Y'}(\varphi x, \varphi y, \varphi z) = \mu_Y(x,y,z), \qquad p_{Y'}(\varphi x) = p_Y(x)$$

for all $x, y, z \in H_Y$. Conversely, every admissible system is realized by such a manifold.

5.1. **The proof of theorem 1.1.**

Proof of theorem 1.1. By proposition 2.6, the manifold X_- is simply connected and has torsion-free integral homology, with

$$H^2(X_-) = \mathbb{Z}\ell, \qquad H^3(X_-) = 0.$$

Thus $r_{X_-} = 0$. The calculations in lemma 4.1 and propositions 4.2 and 4.3 give

$$\mu_{X_-}(\ell,\ell,\ell) = 1, \qquad w_2(X_-) = 0, \qquad p_{X_-}(\ell) = 4.$$

Let $u = c_1(\mathcal{O}_{\mathbb{P}^3}(1))$ be the first Chern class of the standard complex structure on $\mathbb{P}^3$. Then

$$u^3 = 1. \qquad w_2(\mathbb{P}^3) = 0, \qquad p_1(\mathbb{P}^3) = c_1^2 - 2c_2 = 4u^2.$$

Hence X_- and $\mathbb{P}^3$ have isomorphic Wall systems, the isomorphism being $\ell \mapsto u$. By theorem 5.1,

$$X_- \cong_{\mathrm{diff}} \mathbb{P}^3.$$

This complex structure is not the standard one. Indeed, proposition 4.2 gives $c_1(TX_-) = 0$, whereas the standard complex structure on $\mathbb{P}^3$ has first Chern class $4u$.

We next compare it with the construction of Huckleberry, Kebekus and Peternell. Given a complex structure X on S^6 and a point $p \in X$, they consider

$$\pi_p\colon Z_p := \mathrm{Bl}_p X \longrightarrow X$$

and observe that Z_p is diffeomorphic to $\mathbb{P}^3$; their automorphism theorem then yields a one-dimensional family of such complex structures [5, p. 102 and Corollary 1.2]. Let $E_p \cong \mathbb{P}^2$ be the exceptional divisor and put $e_p = [E_p]$. The point-blow-up formula gives

$$H^2(Z_p) = \mathbb{Z}e_p, \qquad e_p^3 = 1,$$

and

$$K_{Z_p} \cong \pi_p^* K_X \otimes \mathcal{O}_{Z_p}(2E_p).$$

Since $H^2(X) = 0$, it follows that

$$c_1(TZ_p) = -2e_p \neq 0.$$

Consequently X_- is not biholomorphic to any of these point blow-ups. Nor are they deformation equivalent through smooth compact complex manifolds since in a smooth proper holomorphic family, the first Chern class of the relative tangent bundle forms a locally constant class. A zero first Chern class therefore cannot deform to $-2e_p$ or $4u$. This proves all the assertions of theorem 1.1. □

Proposition 5.2. *The complex threefold $\widetilde{X}$ is orientation-preservingly diffeomorphic, but not biholomorphic, to the smooth quadric threefold $Q_3 \subset \mathbb{P}^4$.*

Proof. Computing the topology invariants of $\widetilde{X}$, we have $E^3 = 2, w_2(\widetilde{X}) = [E] \bmod 2, \langle p_1(\widetilde{X})[E], [\widetilde{X}]\rangle = 2$. By Wall–Jupp's classification, and comparing to the topology invariants of Q_3, we find that $\widetilde{X} \cong_{\text{diff}} Q_3$.

The first Chern class of $\widetilde{X}$ is

$$c_1(T\widetilde{X}) = -[E],$$

which is primitive in $H^2(\widetilde{X})$. By contrast,

$$c_1(TQ_3) = 3c_1(\mathcal{O}_{Q_3}(1))$$

has divisibility three. A biholomorphism preserves the first Chern class and its divisibility in integral cohomology. Hence $\widetilde{X}$ cannot be biholomorphic to Q_3. □

Remark 5.3. *The point-blow-up construction does not yield a second route to $S^2 \times S^4$. Indeed, for $p \notin \{Q, Q'\}$, let*

$$Z_p := \mathrm{Bl}_p(X_{\text{sph}}).$$

We may choose a double curve $D \subset X_{\text{sph}}$ disjoint from p and perform the Atiyah flop along its strict transform in Z_p. The resulting complex threefold, however, is not diffeomorphic to $S^2 \times S^4$; its underlying smooth manifold is again $\mathbb{P}^3$ (again can be proved by Wall's classification), endowed with an exotic complex structure.

5.2. The proof of theorem 1.2.

Proof of theorem 1.2. By proposition 3.5, X_+ is simply connected and its integral homology is torsion-free, with

$$H^2(X_+) = \mathbb{Z}\ell_+, \qquad H^3(X_+) = 0.$$

Hence $r_{X_+} = 0$. By propositions 4.4 to 4.6,

$$\mu_{X_+}(\ell_+, \ell_+, \ell_+) = 0, \qquad w_2(X_+) = 0, \qquad p_{X_+}(\ell_+) = 0.$$

Let $x = \mathrm{pr}_1^* a$ generate $H^2(S^2 \times S^4)$, and let $y = \mathrm{pr}_2^* b$ generate its fourth cohomology, with

$$\left\langle x \cup y, [S^2 \times S^4] \right\rangle = 1.$$

The Künneth theorem gives

$$x^2 = 0, \qquad H^3(S^2 \times S^4) = 0.$$

Moreover every sphere is stably parallelizable:

$$TS^n \oplus \mathbb{R} \cong \mathbb{R}^{n+1}.$$

Consequently

$$T(S^2 \times S^4) \oplus \mathbb{R}^2 \cong \mathrm{pr}_1^*(TS^2 \oplus \mathbb{R}) \oplus \mathrm{pr}_2^*(TS^4 \oplus \mathbb{R})$$

is trivial. Stable characteristic classes therefore give

$$w_2(S^2 \times S^4) = 0, \qquad p_1(S^2 \times S^4) = 0.$$

Thus $S^2 \times S^4$ has Wall system

$$(0, \mathbb{Z}, 0, 0),$$

and the isomorphism $\ell_+ \mapsto x$ identifies it with the Wall system of X_+. By theorem 5.1, there is an orientation-preserving diffeomorphism

$$X_+ \cong_{\text{diff}} S^2 \times S^4.$$

Since X_+ is a complex threefold by the flop construction in definition 3.2, this proves theorem 1.2. □

6. Further discussion

6.1. **Dependence on the flopping curve.** Retain the labelling of [7, Proposition 4.6(iii)], so that the three cusp double curves correspond to the unoriented directions $e_1, e_2, e_1 - e_2$, and the fixed curve is the one of direction e_2 by [7, Proposition 9.24]. After blowing up the fixed curve and taking the μ_2-quotient, the strict transforms of the other two double curves give disjoint curves

$$C_-, C'_- \subseteq X_-,$$

with normal bundles $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$. Thus either curve could be flopped. The same Wall-invariant calculation shows that both resulting complex manifolds are diffeomorphic to $S^2 \times S^4$.

This local toric symmetry does not by itself extend to the completed global family. The connected automorphism group computed in [7, Proposition 9.23] is the vertical $\mathbb{C}^*$, and its toric action preserves each orbit closure, hence each double curve separately; see the proof of [7, Proposition 9.24]. We therefore do not yet know whether the two resulting complex manifolds are biholomorphic, or even deformation equivalent. We leave this as a natural question for further study.

6.2. **The role of the source construction.** The construction of X_{sph} in [7] is extensive: it develops the period family, the toric cusp filling, the two logarithmic transforms, the integral topology, and a detailed analysis of the automorphism group. Reading and checking the entire construction is correspondingly demanding. Once its existence and diffeomorphism theorems are taken as input, not every one of those calculations is logically required for a direct proof of theorem 1.2.

The additional material is nevertheless geometrically fruitful. In particular, the vertical $\mathbb{C}^*$-action, its fixed curve, the three double curves of the cusp fibre, and the explicit A_2-fan are precisely the data that reveal the quotient resolution and the floppable curve used here. These features are invisible in the bare statement that S^6 carries a complex structure. Thus the breadth of the source construction should not be regarded merely as redundancy: its more exploratory calculations expose birational geometry that suggests new constructions. It would be interesting to determine whether other parts of that geometry lead to further compact complex manifolds or to new complex structures on familiar smooth six-manifolds.

Recently, the main result of [7] has been claimed to be formalized by Lean 4 by Boris Alexeev. See https://github.com/plby/HopfProblem.

References

[1] M. F. Atiyah. On analytic surfaces with double points. *Proc. Roy. Soc. London Ser. A*, 247(1249):237–244, 1958.
[2] A. Borel and J.-P. Serre. Groupes de Lie et puissances réduites de Steenrod. *Amer. J. Math.*, 75(3):409–448, 1953.
[3] C. Ehresmann. Sur les variétés presque complexes. In *Proceedings of the International Congress of Mathematicians, Cambridge, Massachusetts, 1950*, volume 2, pages 412–419, Providence, RI, 1952. American Mathematical Society.
[4] H. Hopf. Zur Topologie der komplexen Mannigfaltigkeiten. In *Studies and Essays Presented to Richard Courant on His 60th Birthday, January 8, 1948*, pages 167–185. Interscience Publishers, Inc., New York, NY, 1948.
[5] A. T. Huckleberry, S. Kebekus, and T. Peternell. Group actions on S^6 and complex structures on $\mathbb{P}^3$. *Duke Math. J.*, 102(1):101–124, 2000.
[6] P. E. Jupp. Classification of certain 6-manifolds. *Proc. Cambridge Philos. Soc.*, 73(2):293–300, 1973.
[7] A compact complex threefold fibred by tori over the projective line, and the six-sphere, 2026. Unpublished anonymous preprint.
[8] W. A. Sutherland. A note on almost complex and weakly complex structures. *J. London Math. Soc.*, 40:705–712, 1965.
[9] C. T. C. Wall. Classification problems in differential topology. V. On certain 6-manifolds. *Invent. Math.*, 1:355–374, 1966. Corrigendum: Invent. Math. 2 (1966), 306.

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